Prompt 1 — Create the basic application

Create an interactive educational web application titled **"Continuous Probability Distributions: Industry Applications"** for undergraduate AIML students.

Build a clean, responsive interface with a home page and three modules:

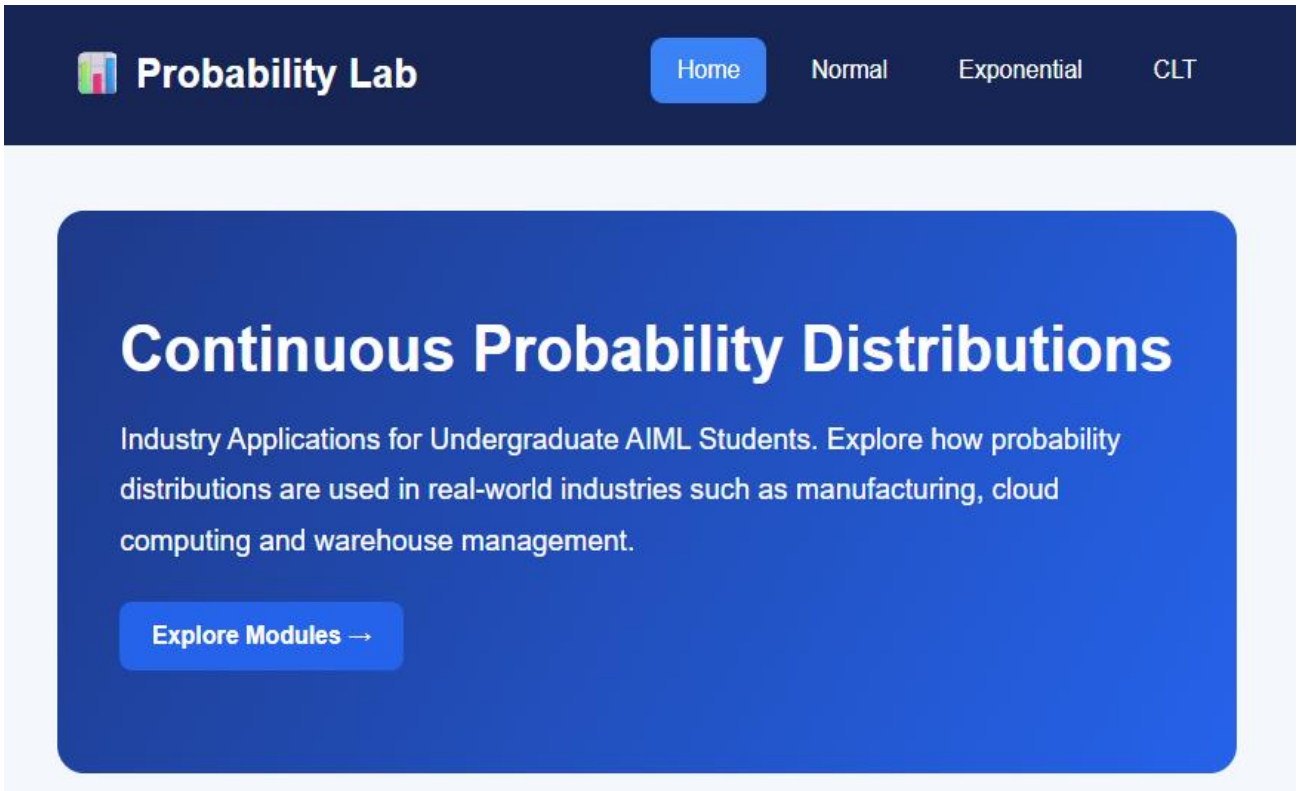
1. Normal Distribution — Manufacturing Quality Control
2. Exponential Distribution — Cloud Request Inter-arrival Time
3. Central Limit Theorem — Warehouse Processing Time

Use simple language suitable for beginners. Each module must include:

- Industry problem statement
- Input parameters
- Mathematical formula
- Interactive calculation section
- Dynamic visualization
- Interpretation of results

Use Python only if the chosen framework supports it; otherwise use the framework's native numerical and charting libraries. Implement the application using the existing project setup and do not introduce unnecessary dependencies.

For now, create the navigation and page layouts. Do not implement all calculations yet.



Learning Modules

Normal Distribution

Understand manufacturing quality control and how product measurements follow a normal distribution.

Open Module

Exponential Distribution

Explore cloud request arrival times and understand the waiting time between incoming requests.

Open Module

Central Limit Theorem

Learn how warehouse processing times can be analyzed using sample means and the Central Limit Theorem.

Open Module

Learning Objectives

- ✓ Understand continuous probability distributions.
- ✓ Connect mathematical concepts with industry applications.
- ✓ Explore distribution parameters and mathematical formulas.
- ✓ Interpret graphs and probability results.

Normal Distribution

Industry Application: Manufacturing Quality Control

1. Industry Problem Statement

A manufacturing company produces metal components with a target diameter. Small variations occur during production. The quality control department wants to understand whether the manufactured products meet the required diameter specifications. Normal distribution helps analyze these variations and identify products that may be defective.

2. Input Parameters

Mean (μ)

Enter mean

Standard Deviation (σ)

Enter standard deviation

Product Diameter (x)

Enter diameter

Lower Specification Limit

Enter lower limit

Upper Specification Limit

Enter upper limit

3. Mathematical Formula

Probability Density Function (PDF):

$$f(x) = 1 / (\sigma\sqrt{2\pi}) \times e^{-(x-\mu)^2/2\sigma^2}$$

Z-score:

$$Z = (x - \mu) / \sigma$$

4. Interactive Calculation

Enter the parameters above to explore the calculation.

Calculate (Coming Soon)

5. Distribution Visualization

Normal Distribution Bell Curve will appear here.

6. Interpretation of Results

Probability results and product quality interpretation will appear here.

Prompt 2 — Implement the Normal Distribution module

Implement the **Normal Distribution — Manufacturing Quality Control** module.

Use this illustrative example:

- Target shaft diameter: 10 mm
- Mean diameter μ = 10 mm
- Standard deviation σ = 0.05 mm
- Lower specification limit = 9.90 mm
- Upper specification limit = 10.10 mm

Provide interactive inputs for μ, σ, lower limit, and upper limit.

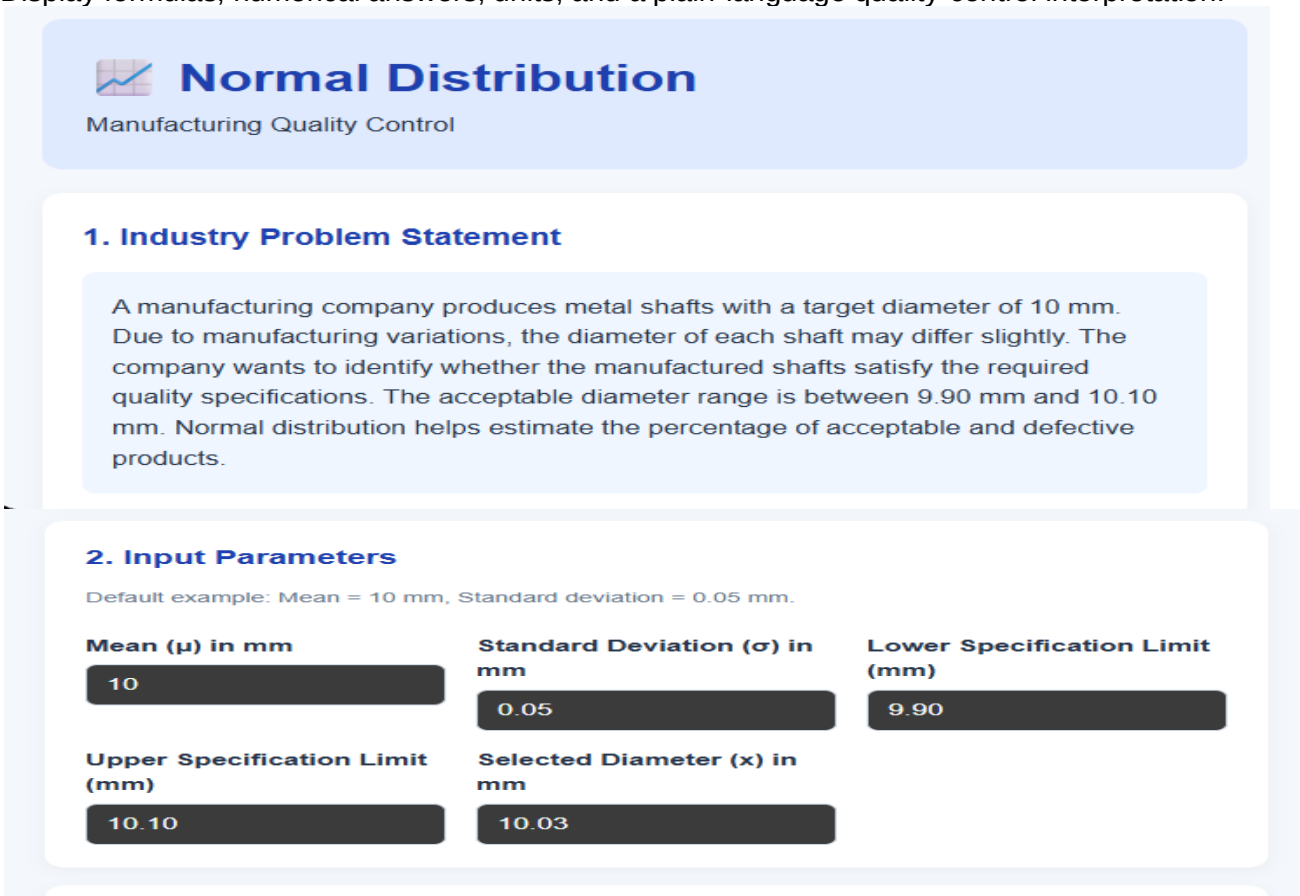
Calculate:

1. Probability density at a selected diameter.
2. Probability that diameter lies between the lower and upper specification limits.
3. Probability of a diameter below the lower limit.
4. Probability of a diameter above the upper limit.
5. Expected percentage of defective products.

Use the Normal CDF to calculate interval probabilities. Display a dynamic bell-shaped density curve and shade the acceptable region and the two defective regions in different colors.

When any input changes, update the calculations and graph immediately. Validate that σ is positive and the lower limit is less than the upper limit.

Display formulas, numerical answers, units, and a plain-language quality-control interpretation.



3. Mathematical Formulas

Probability Density Function (PDF)

$$f(x) = 1 / (\sigma\sqrt{2\pi}) \times e^{-(x-\mu)^2 / 2\sigma^2}$$

Normal Cumulative Distribution Function (CDF)

$$F(x) = P(X \leq x) = \Phi((x-\mu)/\sigma)$$

Probability within specification limits

$$P(L \leq X \leq U) = F(U) - F(L)$$

Probability below lower limit

$$P(X < L) = F(L)$$

Probability above upper limit

$$P(X > U) = 1 - F(U)$$

Expected defective percentage

$$\text{Defective \%} = [P(X < L) + P(X > U)] \times 100$$

4. Interactive Calculations

Results automatically update when you change any input.

PDF at selected diameter

6.664492

mm^{-1}

Acceptable Products

95.4500%

0.954500 probability

Below Lower Limit

2.2750%

0.022750 probability

Above Upper Limit

2.2750%

0.022750 probability

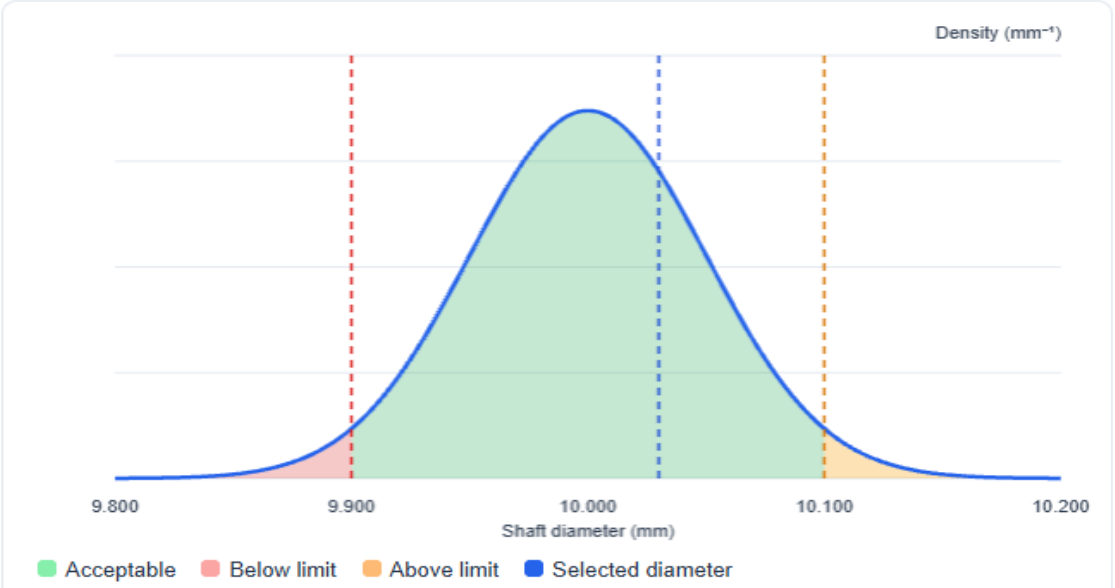
Expected Defective Products

4.5500%

Percentage

5. Dynamic Normal Distribution Graph

Green = Acceptable products. Red = Below lower specification. Orange = Above upper specification. Blue dashed line = Selected diameter.



6. Interpretation of Results

Acceptable products: 95.4500%.
Defective products: 4.5500%.
Below lower limit: 2.2750%.
Above upper limit: 2.2750%.

The estimated defective proportion is moderate. The manufacturing process should be monitored.

At diameter 10.0300 mm, the probability density is 6.664492 mm^{-1}. This is a density value, not the probability of one exact diameter.

Prompt 3 — Add an interactive Exponential Distribution module

Implement the Exponential Distribution — Cloud Request Inter-arrival Time module.

Use an illustrative Poisson request-arrival rate of $\lambda = 12$ requests per minute.

Allow students to change:

- Arrival rate λ in requests per minute
- Time threshold t in seconds

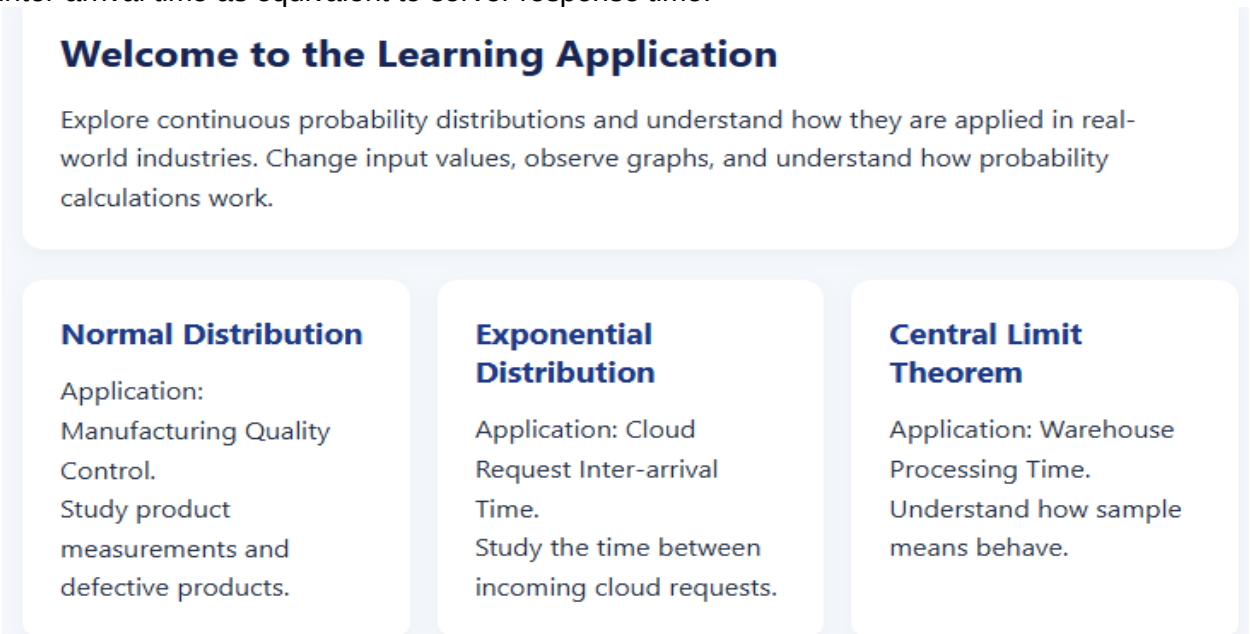
Calculate:

1. Mean inter-arrival time = $1/\lambda$.
2. Variance of inter-arrival time = $1/\lambda^2$, using consistent units.
3. $P(T \leq t) = 1 - \exp(-\lambda t)$.
4. $P(T > t) = \exp(-\lambda t)$.
5. $P(T > t_2 \mid T > t_1)$, where $t_2 \geq t_1$.

Convert seconds and minutes correctly before calculations. Display the exponential probability density curve and shade the relevant probability region.

Explain the memoryless property in simple language. Validate all inputs and show units beside every result.

Clearly state that the model assumes a constant-rate Poisson arrival process. Do not describe inter-arrival time as equivalent to server response time.



Continuous Probability Distributions

Understanding Industry Applications Through Interactive Learning

Home | Normal Distribution | Exponential Distribution | Central Limit Theorem

Exponential Distribution — Cloud Request Inter-arrival Time

Industry Problem

A cloud service receives requests from users. We want to understand the time between consecutive incoming requests. The Exponential Distribution helps estimate the probability that the next request arrives within a particular time.

Illustrative arrival rate (λ) = 12 requests per minute

Time between requests = Inter-arrival time

Distribution = Exponential Distribution

Important assumption: This model assumes a constant-rate Poisson request-arrival process. Inter-arrival time means the time between incoming requests. It is not the same as server response time.

Change Input Values

Arrival Rate λ (requests per minute): 12

Time Threshold t (seconds): 5

First Time t_1 (seconds): 5

Second Time t_2 (seconds): 10

You can change all values to explore different arrival rates and time thresholds. Calculations update automatically.

Exponential Distribution Calculations

Mean inter-arrival time: **0.083333 minutes (5.000000 seconds)**

Variance of inter-arrival time: **0.006944 minutes² (25.000000 seconds²)**

$P(T \le t)$: **63.2121%**

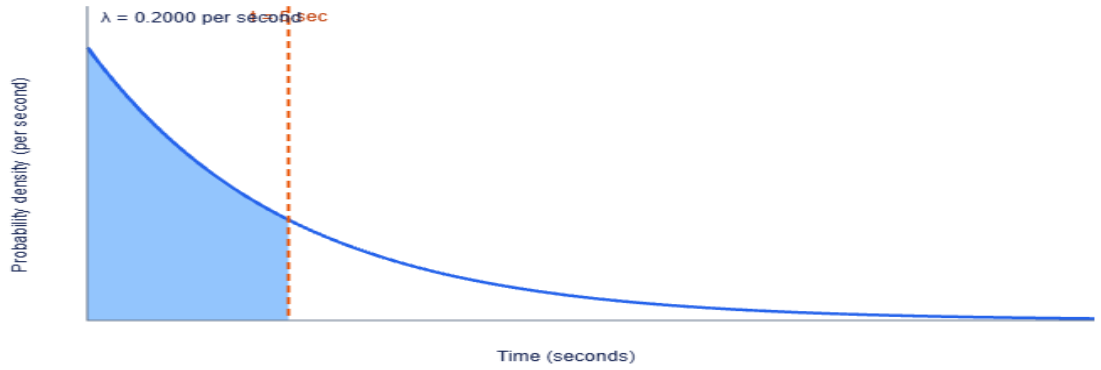
$P(T > t)$: **36.7879%**

$P(T > t_2 \mid T > t_1)$: **36.7879%**

PDF at time t: **0.073576 per second**

At an arrival rate of 12 requests per minute, the probability that the next request arrives within 5 seconds is 63.2121%. The probability that it takes longer than 5 seconds is 36.7879%.

Exponential Probability Density Curve



The shaded region represents P(T ≤ t), the probability that the next request arrives within the selected time.

The shaded region represents P(T ≤ t), the probability that the next request arrives within the selected time.

Graph explanation: The curve starts at its highest value and decreases as time increases. The shaded area from 0 to t represents the probability of an inter-arrival time less than or equal to t.

Formulas Used

Mean = $1 / \lambda$

Variance = $1 / \lambda^2$

$P(T \le t) = 1 - \exp(-\lambda t)$

$P(T > t) = \exp(-\lambda t)$

$P(T > t_2 \mid T > t_1) = \exp(-\lambda(t_2 - t_1))$

PDF: $f(t) = \lambda \exp(-\lambda t)$, for $t \ge 0$

Since the arrival rate is entered in requests per minute and time is entered in seconds, the application converts the rate into requests per second before performing probability calculations.

Understanding the Memoryless Property

The Exponential Distribution has a special property called the **memoryless property**.
In simple words, the time already waited does not change the probability distribution of the additional waiting time.
For example, suppose no new cloud request has arrived during the first 5 seconds. The probability of waiting more than another 5 seconds is the same as the probability of waiting more than 5 seconds from the beginning.

$$P(T > t_2 \mid T > t_1) = P(T > t_2 - t_1)$$

This property applies under the constant-rate Poisson arrival assumption.